MIL-M-28787/54B(EC)
19 July 1982
SUPERSEDING
MIL-M-28787/54A(NAVY)
11 December 1975

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, INTERFACE, DIGITAL, KEY CODE QDH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface receiver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.29 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QDH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 40 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

*3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/54-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc and (V_{EE}) of -15 ±2 Vdc when driving the specified load in either logic state, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

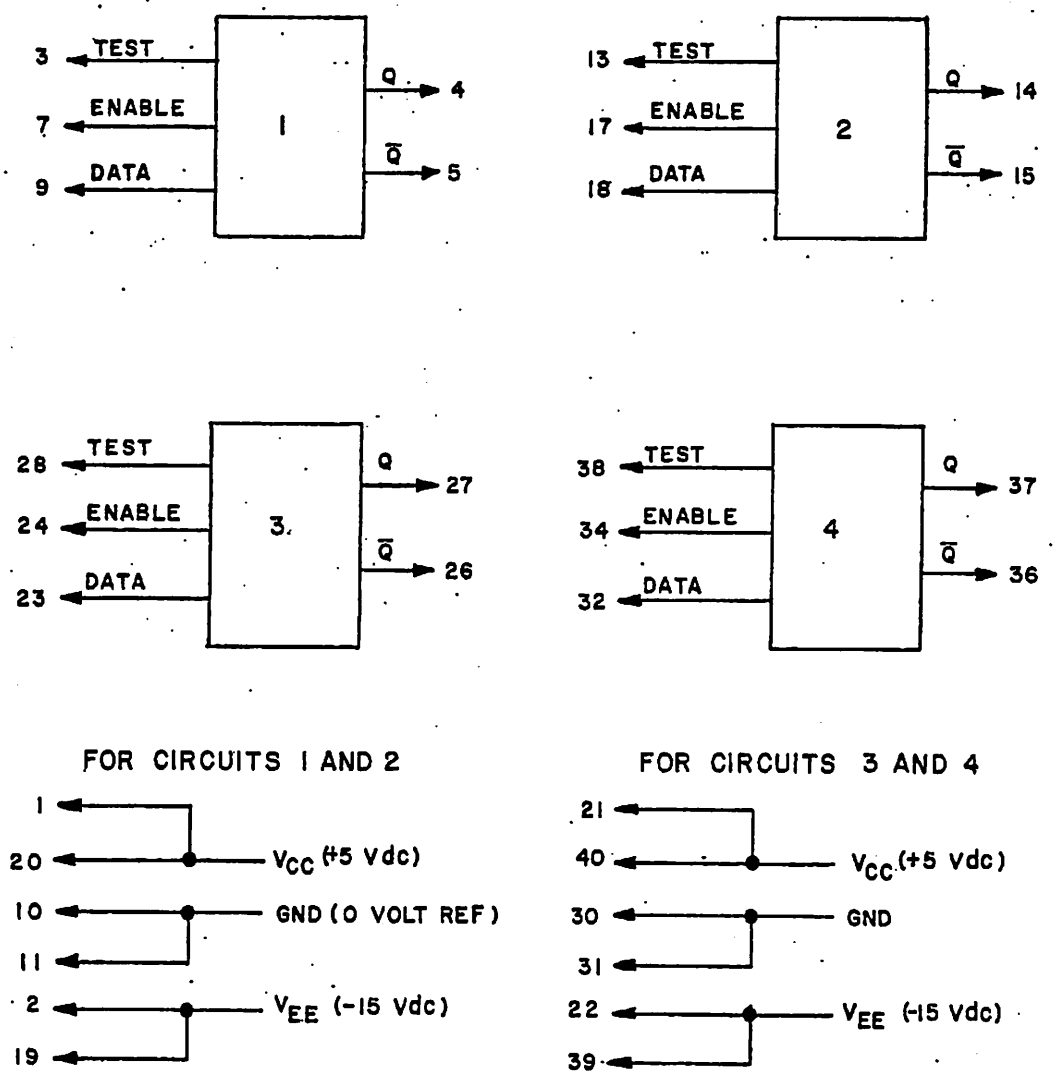
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

*4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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Unused pins - 6, 8, 12, 16, 25, 29, 33, 35

FIGURE 1. QDH module function diagram, interface receiver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	V_{OUT} (low loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1,B2	Isolation	IV	S	10	-	1	-	10	-	1	-	MΩ
C1-C4	Input breakdown current	V	S	0.0	5.0	0.0	5.0	0.0	10.0	0.0	10.0	µA
C5-C8	Input breakdown current	V	S	0.0	15.0	0.0	15.0	0.0	15.0	0.0	15.0	µA
D1,D2	Continuity	VI	S	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
D3-D6	Continuity	VI	S	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	Ω
E1-E4	Input current, low level	VII	X	-0.1	-1.6	-0.1	-1.6	-0.1	-1.6	-0.1	-1.6	mA
E5-E8	Input current, low level	VII	X	-0.2	-3.2	-0.2	-3.2	-0.2	-3.2	-0.2	-3.2	mA
E9-E12	Input current, low level	VII	X	-0.1	-1.0	-0.1	-1.0	-0.1	-1.0	-0.1	-1.0	mA
F1-F4	Input current, high level	VII	X	0.0	2.0	0.0	2.0	0.0	4.0	0.0	5.0	µA
F5-F8	Input current, high level	VII	X	0.0	4.0	0.0	4.0	0.0	8.0	0.0	10.0	µA
F9-F12	Input current, high level	VII	X	0.5	4.0	0.5	4.0	0.5	4.0	0.5	4.0	mA
G1-G4	Output short circuit current	VII	X	-1.3	-4.28	-1.3	-4.28	-1.3	-4.28	-1.3	-4.28	mA
G5-G8	Output short circuit current	VII	X	-0.6	-1.34	-0.6	-1.34	-0.6	-1.34	-0.6	-1.34	mA
H1-H12	Output voltage, high level	VIII	X	3.8	4.5	3.8	4.5	3.8	4.5	3.8	4.5	V
J1-J8	Output voltage, low level	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
K	Power supply current, 5 V	IX	S	30	70	30	70	30	70	30	70	mA
L	Power supply current, -15 V	IX	S	-5	-20	-5	-20	-5	-20	-5	-20	mA
M1-M8	Transition time (t_{TLH})	X	S	50	1,000	50	1,000	50	1,000	50	1,000	ns
N1-N8	Transition time (t_{THL})	X	S	5	25	5	25	5	25	5	25	ns
P1-P4, P9-P12	Propagation delay time (t_{PHL})	X	S	6	40	6	40	6	40	6	40	ns
P5-P8, P13-P16	Propagation delay time (t_{PHL})	X	S	10	110	10	110	10	110	10	110	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q4, Q9-Q12	Propagation delay time (t_{PLH})	X	S	10	80	10	80	10	80	10	80	ns
Q5-Q8, Q13-Q16	Propagation delay time (t_{PLH})	X	S	10	110	10	110	10	110	10	110	ns
R1-R8	Propagation delay time (t_{PHL})	XI	S	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0	µs
S1-S8	Propagation delay time (t_{PLH})	XI	S	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0	µs
T1-T4	Maximum frequency check (measuring voltage level)	XII	S	3.4	5.0	3.4	5.0	3.4	5.0	3.4	5.0	V
U1-U4	Noise rejection, low level	XII	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
V1-V4	Noise rejection, high level	XII	Q	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
W1-W4	Input capacitance	XIII	Q	2	15	2	15	2	15	2	15	pF
W5-W8	Input capacitance	XIII	Q	6	30	6	30	6	30	6	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

*6.3 Cross reference of classification. Part number M28787/54-1 supersedes part number M28787/54. The two parts are interchangeable.

*6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	725 mW (outputs not loaded)
Supply voltage	
V_{CC}	5.0 ±0.5 Vdc
V_{EE}	-15.0 ±2.0 Vdc
Input voltage (test and enable)	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum

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Input voltage (data)	
Low level	-17.5 V minimum, -7.5 V maximum
High level	-4.5 minimum, 1.5 V maximum
Input current (low level)	
Test (V_{IL} = 0.0 V)	-1.6 mA maximum
Enable (V_{IL} = 0.0 V)	-3.2 mA maximum
Data (V_{IL} = -12 V)	-1.0 mA maximum
Input current (high level)	
Test (V_{IH} = 4.0 V)	5.0 µA maximum
(V_{IH} = 5.5 V)	10.0 µA maximum
Enable (V_{IH} = 4.0 V)	10.0 µA maximum
(V_{IH} = 5.5 V)	15.0 µA maximum
Data (V_{IH} = 0.0 V)	4.0 mA maximum
Output voltage	
Low level	0.0 V minimum, 0.40 V maximum
High level	3.8 V minimum, V_{CC} maximum
Output drive current (Q output)	
Low level (V_{OL} = 0.40 V)	9.6 mA maximum
High level (V_{OH} = 3.8 V)	-60 µA maximum
Output drive current ($\overline{Q}$ output)	
Low level (V_{OL} = 0.40 V)	11.2 mA maximum
High level (V_{OH} = 3.8 V)	-60 µA maximum
Propagation delay time (test, Q output)	
t_{PLH}	80 ns maximum
t_{PHL}	40 ns maximum
Propagation delay time (test, $\overline{Q}$ output)	
t_{PLH}	110 ns maximum
t_{PHL}	110 ns maximum
Propagation delay time (enable)	
t_{PLH} (Q output)	80 ns maximum
t_{PLH} ($\overline{Q}$ output)	110 ns maximum
t_{PHL} (Q output)	40 ns maximum
t_{PHL} ($\overline{Q}$ output)	110 ns maximum
Propagation delay time (data, Q or $\overline{Q}$ output)	
t_{PLH}	1.0 µs minimum, 3.0 µs maximum
t_{PHL}	1.0 µs minimum, 3.0 µs maximum

*6.5 Module function. This module contains four independent NTDS (Naval Technical Data Systems) slow line receiver circuits. Each circuit is capable of converting -15/0 volt signals to 0/+5 volt signals which are compatable with the input requirements of typical logic circuits.

*6.6 Input to output relationship. Each interface receiver has the input to output relationship of table II.

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TABLE II. Input to output relationship. 1/-3/

Inputs			Outputs	
Test	Enable	Data	Q	$\overline{Q}$
X	H	H	H	L
X	H	L	L	H
X	H	Open	L	H
H	L	X	L	H
L	L	X	H	L

1/ H = High-voltage level.
2/ L = Low-voltage level.
3/ X = Don't care.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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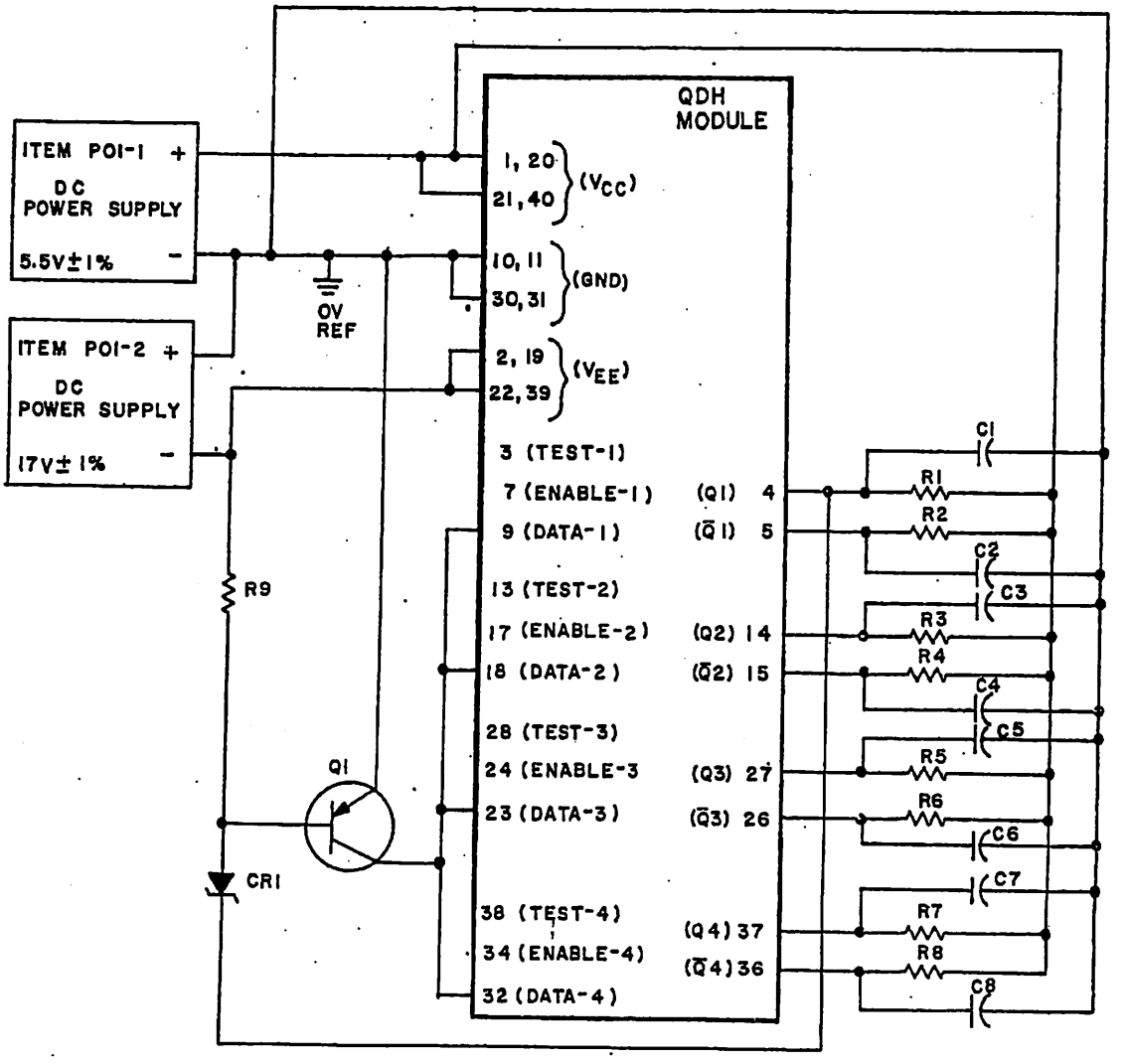
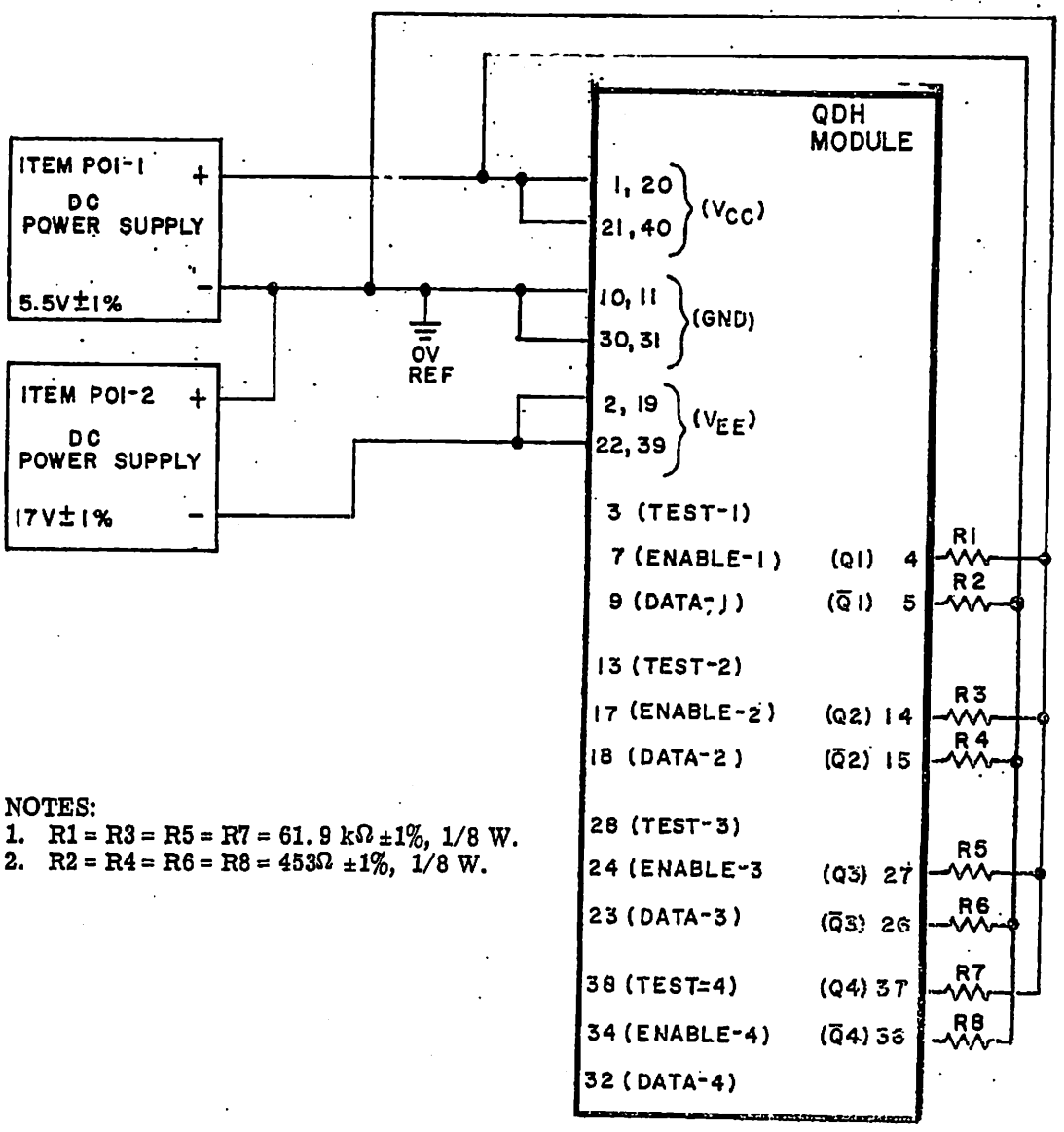


FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R3 = R5 = R7 = 61.9 kΩ ±1%, 1/8 W.
2. R2 = R4 = R6 = R8 = 453Ω ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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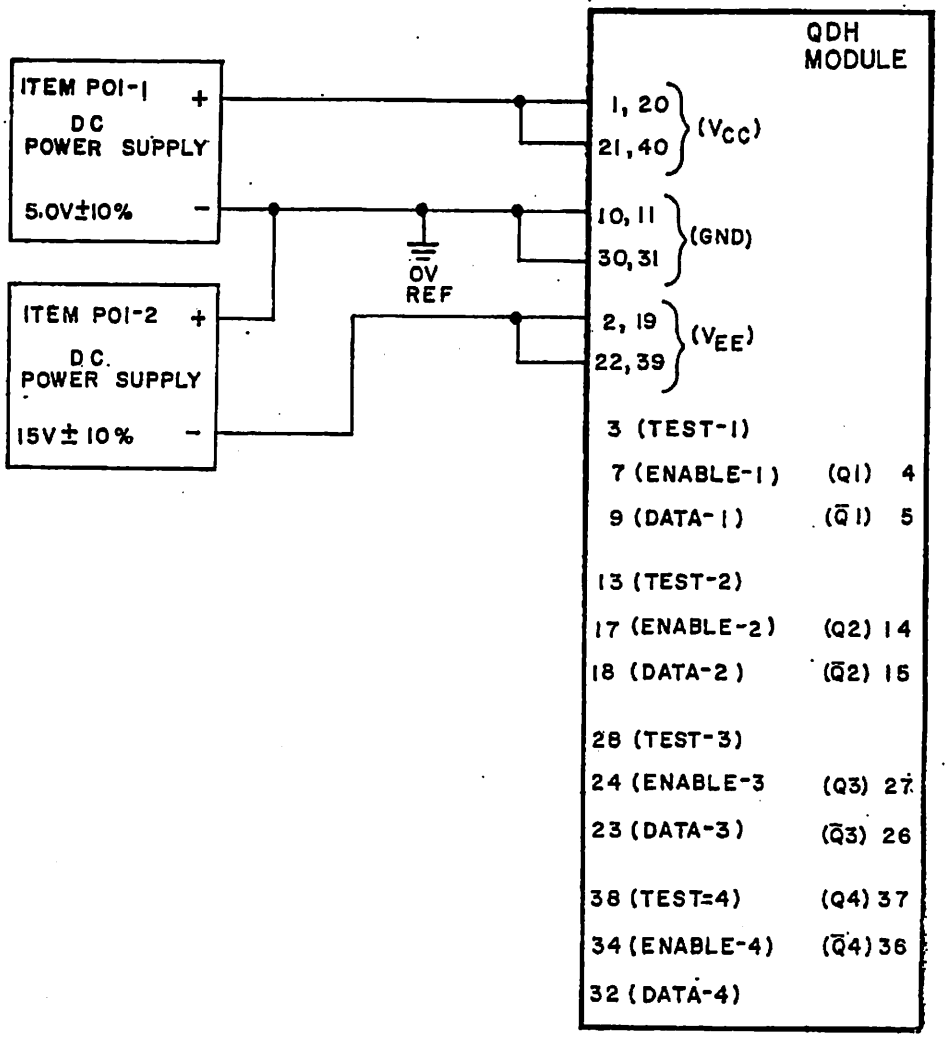


FIGURE 4. Vibration test circuit.

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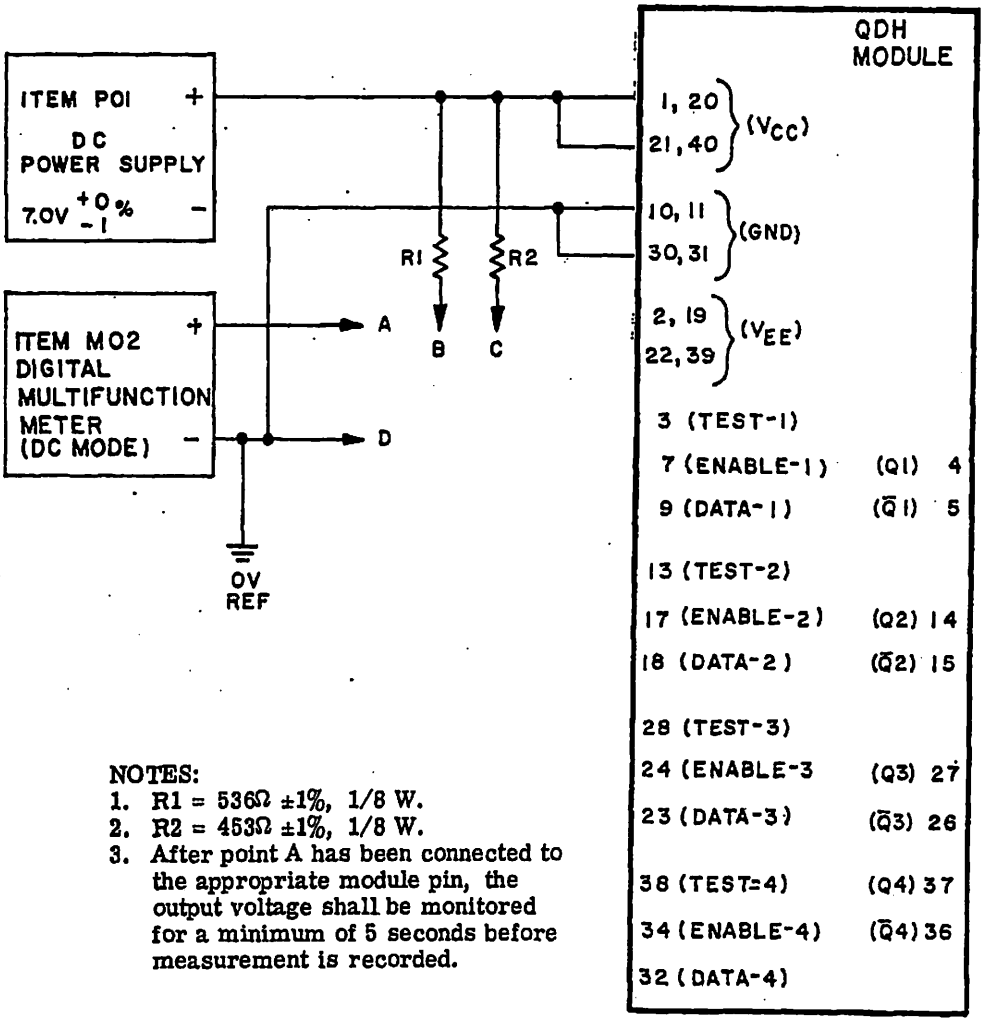
TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
A1	Q1 V_{OUT} (low loaded)	4	4	Open	7	0.1	0.8	V	5
A2	$\overline{Q1}$	5	Open	5	3,7				
A3	Q2	14	14	Open	17				
A4	$\overline{Q2}$	15	Open	15	13,17				
A5	Q3	27	27	Open	24				
A6	$\overline{Q3}$	26	Open	26	24,28				
A7	Q4	37	37	Open	34				
A8	$\overline{Q4}$	36	Open	36	34,38				

*TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation	6,8,12,16,25, 29,33,35	All other pins	10	-	MΩ	6
B2	Isolation	All pins	Frame	10	-	MΩ	6

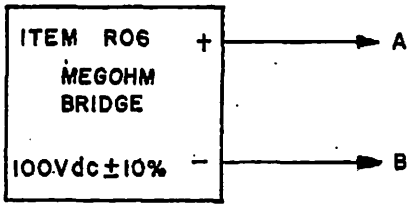
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NOTES:
1. R1 = 536Ω ±1%, 1/8 W.
2. R2 = 453Ω ±1%, 1/8 W.
3. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before measurement is recorded.

FIGURE 5. V_{out} (low-loaded) test circuit for table III.

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition B2 (when measuring to frame) penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

	QDH MODULE
1, 20, 21, 40 (V_{CC})	
10, 11, 30, 31 (GND)	
2, 19, 22, 39 (V_{EE})	
3 (TEST-1)	
7 (ENABLE-1)	(Q1) 4
9 (DATA-1)	($\bar{Q}$1) 5
13 (TEST-2)	
17 (ENABLE-2)	(Q2) 14
18 (DATA-2)	($\bar{Q}$2) 15
28 (TEST-3)	
24 (ENABLE-3	(Q3) 27
23 (DATA-3)	($\bar{Q}$3) 26
38 (TEST=4)	(Q4) 37
34 (ENABLE-4)	($\bar{Q}$4) 36
32 (DATA-4)	

FIGURE 6. Isolation test circuit for table IV.

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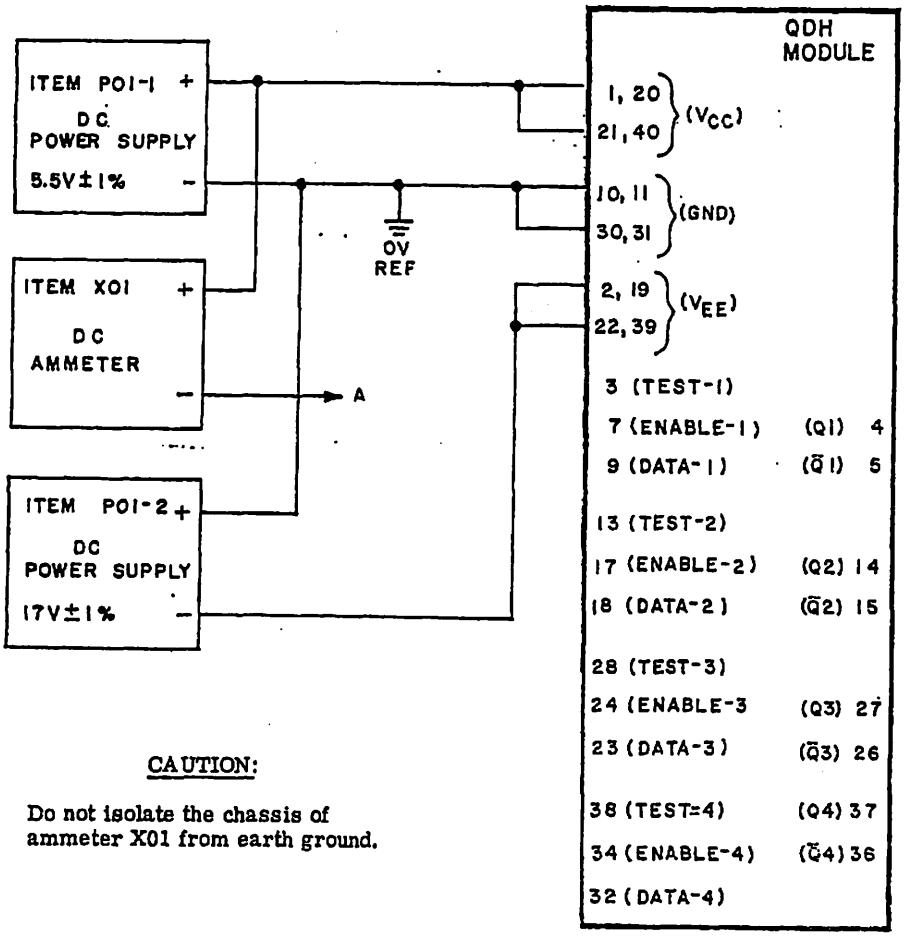
TABLE V. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	TEST-1 Input breakdown current	3	0.0	5.0	µA	7
C2	TEST-2	13				
C3	TEST-3	28				
C4	TEST-4	38				
C5	ENABLE-1	7		15.0		
C6	ENABLE-2	17				
C7	ENABLE-3	24				
C8	ENABLE-4	34				

TABLE VI. Continuity test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	Continuity	10	11	0.0	0.1	Ω	8
D2		30	31		0.1		
D3		1	20		1.5		
D4		2	19				
D5		21	40				
D6		22	39				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 7. Input breakdown current test circuit for table V.

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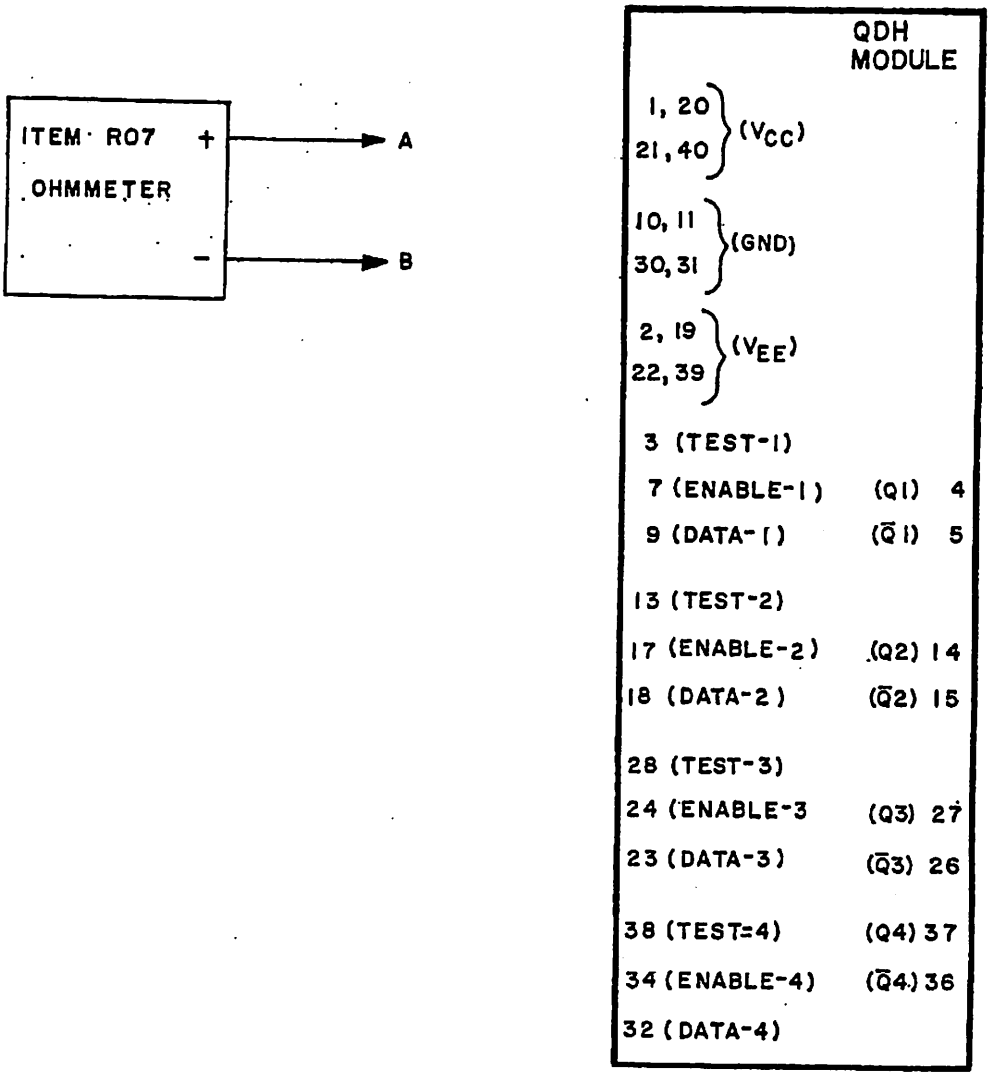


FIGURE 8. Continuity test circuit for table VI.

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TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	TEST-1	F	Open	Open	Open	3	A	-0.1	-1.6	mA	9
E2	TEST-2					13					
E3	TEST-3					28					
E4	TEST-4					38					
E5	ENABLE-1					7		-0.2	-3.2		
E6	ENABLE-2					17					
E7	ENABLE-3					24					
E8	ENABLE-4					34					
E9	DATA-1	Open	F			9	B	-0.1	-1.0		
E10	DATA-2					18					
E11	DATA-3					23					
E12	DATA-4					32					

Input current, low level

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

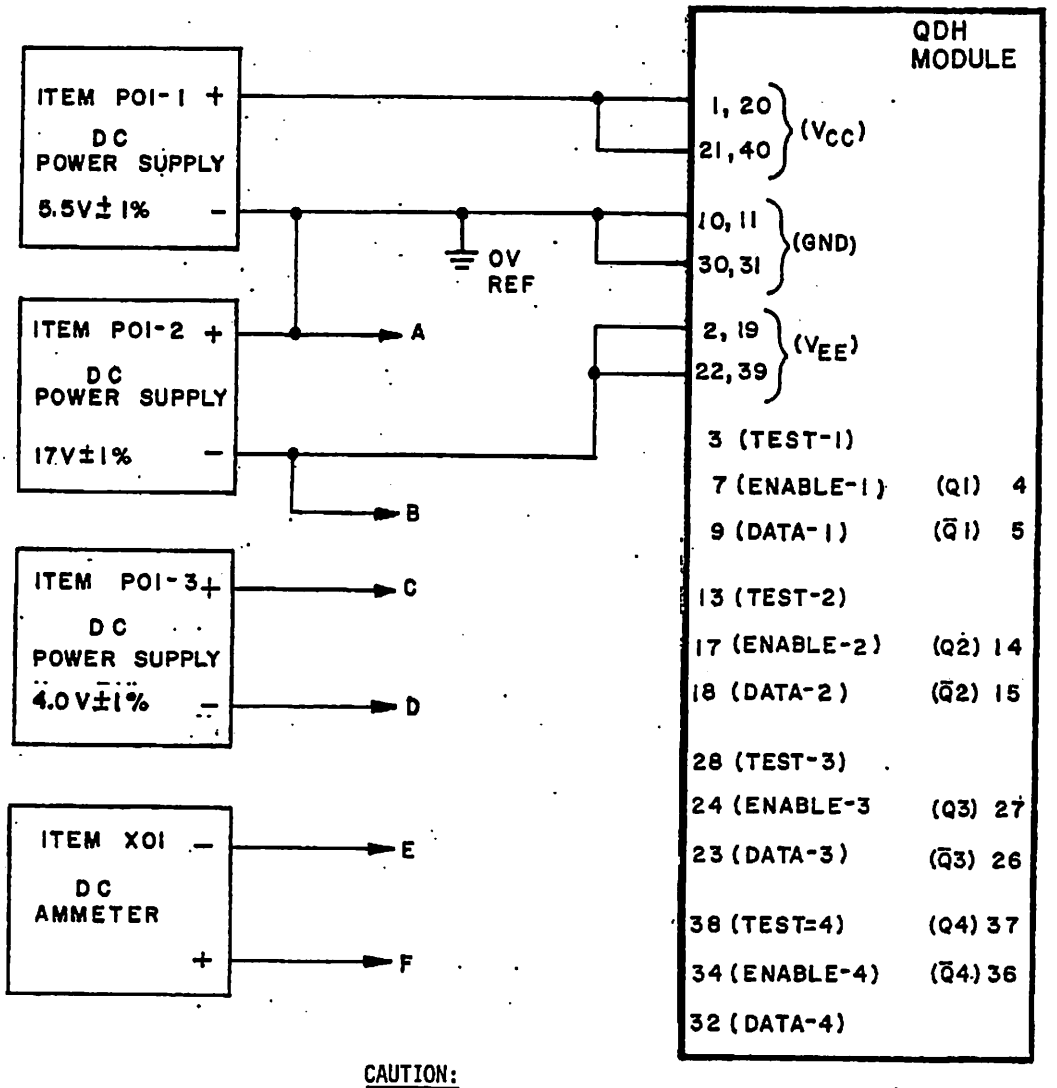
Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	TEST-1 Input current, high level	7,9,13,17, 18,23,24,28, 32,34,38,F	Open	3	E	D	A	0.0	2.0	µA	9
F2	TEST-2	7,9,17,18, 23,24,28,32, 34,38,F		13							
F3	TEST-3	7,9,17,18, 23,24,32,34, 38,F		28							
F4	TEST-4	7,9,17,18, 23,24,32,34, F		38							
F5	ENABLE-1	9,17,18,23, 24,32,34,F		7					4.0		
F6	ENABLE-2	9,18,23,24, 32,34,F		17							
F7	ENABLE-3	9,18,23,32, 34,F		24							
F8	ENABLE-4	9,18,23,32, F		34							
F9	DATA-1	18,23,32,F		Open	Open	9		0.5			
F10	DATA-2	23,32,F				18					
F11	DATA-3	32,F				23					
F12	DATA-4	F				32					

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G1	Q1 Output short circuit current	9, F	Open	Open	Open	4	A	-1.3	-4.28	mA	9
G2	Q2	18, F				14					
G3	Q3	23, F				27					
G4	Q4	32, F				37					
G5	$\overline{Q}$1	7, F, D		3	A	5		-0.6	-1.34		
G6	$\overline{Q}$2	17, F, D		13		15					
G7	$\overline{Q}$3	24, F, D		28		26					
G8	$\overline{Q}$4	34, F, D		38		36					

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: When performing short circuit current, only one output shall be shorted at a time and the measurement taken within 10 seconds.

FIGURE 9. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H1	Q1	Output voltage, high level	Open	4	E	7	C	9	3.8	4.5	V	10
H2	Q2			14		17		18				
H3	Q3			27		24		23				
H4	Q4			37		34		32				
H5	$\overline{Q1}$			5	E,3	7		9				
H6	$\overline{Q2}$			15	E,13	17		18				
H7	$\overline{Q3}$			26	E,28	24		23				
H8	$\overline{Q4}$			36	E,38	34		32				
H9	Q1			4	Open	3,7	Open	Open				
H10	Q2			14		13,17						
H11	Q3			27		24,28						
H12	Q4			37		34,38						
J1	Q1	Output voltage, low level		4	F,9	7	3	C	0.0	0.4		
J2	Q2			14	F,18	17	13					
J3	Q3			27	F,23	24	28					
J4	Q4			37	F,32	34	38					
J5	$\overline{Q1}$		9	5	Open	3,7	Open	Open				
J6	$\overline{Q2}$		18	15		13,17						
J7	$\overline{Q3}$		23	26		24,28						
J8	$\overline{Q4}$		32	36		34,38						

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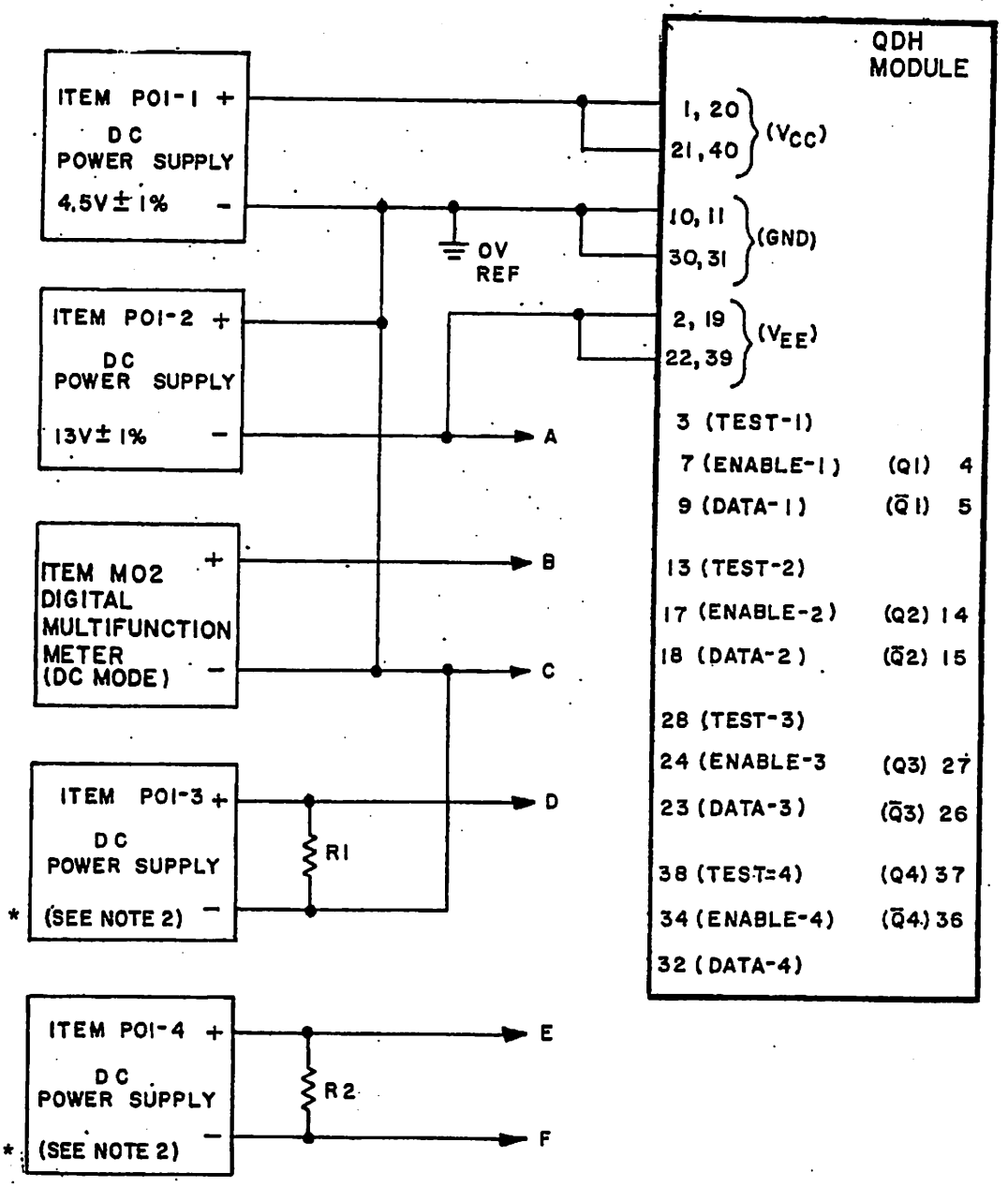


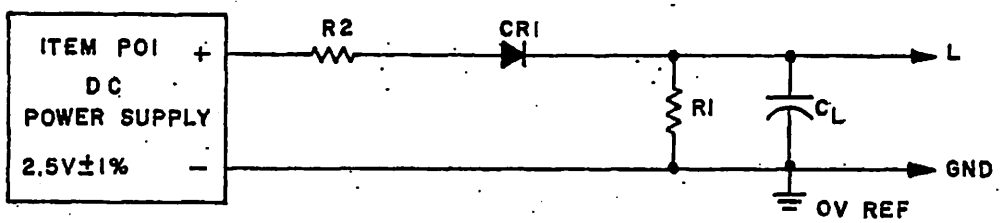
FIGURE 10. Output voltage (high-level and low-level) test circuit for table VIII.

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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
* 2. For test conditions H1-H4, item PO1-3 = 2.0 V ±1% and item PO1-4 = 4.5 V ±1%.
For test conditions H5-H8, item PO1-3 = 2.0 V ±1% and item PO1-4 = 7.5 V ±1%.
For test conditions H9-H12, item PO1-3 = 0.8 V ±1%.
For test conditions J1-J4, item PO1-3 = 0.8 V ±1% and item PO1-4 = 2.0 V ±1%.
For test conditions J5-J8, item PO1-3 = 0.8 V ±1%.
3. Load the outputs of the circuit under test with the loads as specified on figure 11.

FIGURE 10. Output voltage (high-level and low-level) test circuit for table VIII - Continued.

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NOTES:
Load 1 for Q outputs:
1. CR1 = 1N4150, or equivalent.
2. C_L = 40 ±5 pF (including probe and parasitic capacitance).
3. R1 = 61.9 kΩ ±1%, 1/8 W.
4. R2 = Select value for current flow of 9.6 mA ±5% when point L is held at 0.4 V.
5. C_L is not required for test conditions H and J of table VIII.

Load 2 for $\bar{Q}$ outputs:
1. CR1 = 1N4150, or equivalent.
2. C_L = 40 ±5 pF (including probe and parasitic capacitance).
3. R1 = 61.9 kΩ ±1%, 1/8 W.
4. R2 = Select value for current flow of 11.2 mA ±5% when point L is held at 0.4 V.
5. C_L is not required for test conditions H and J of table VIII.

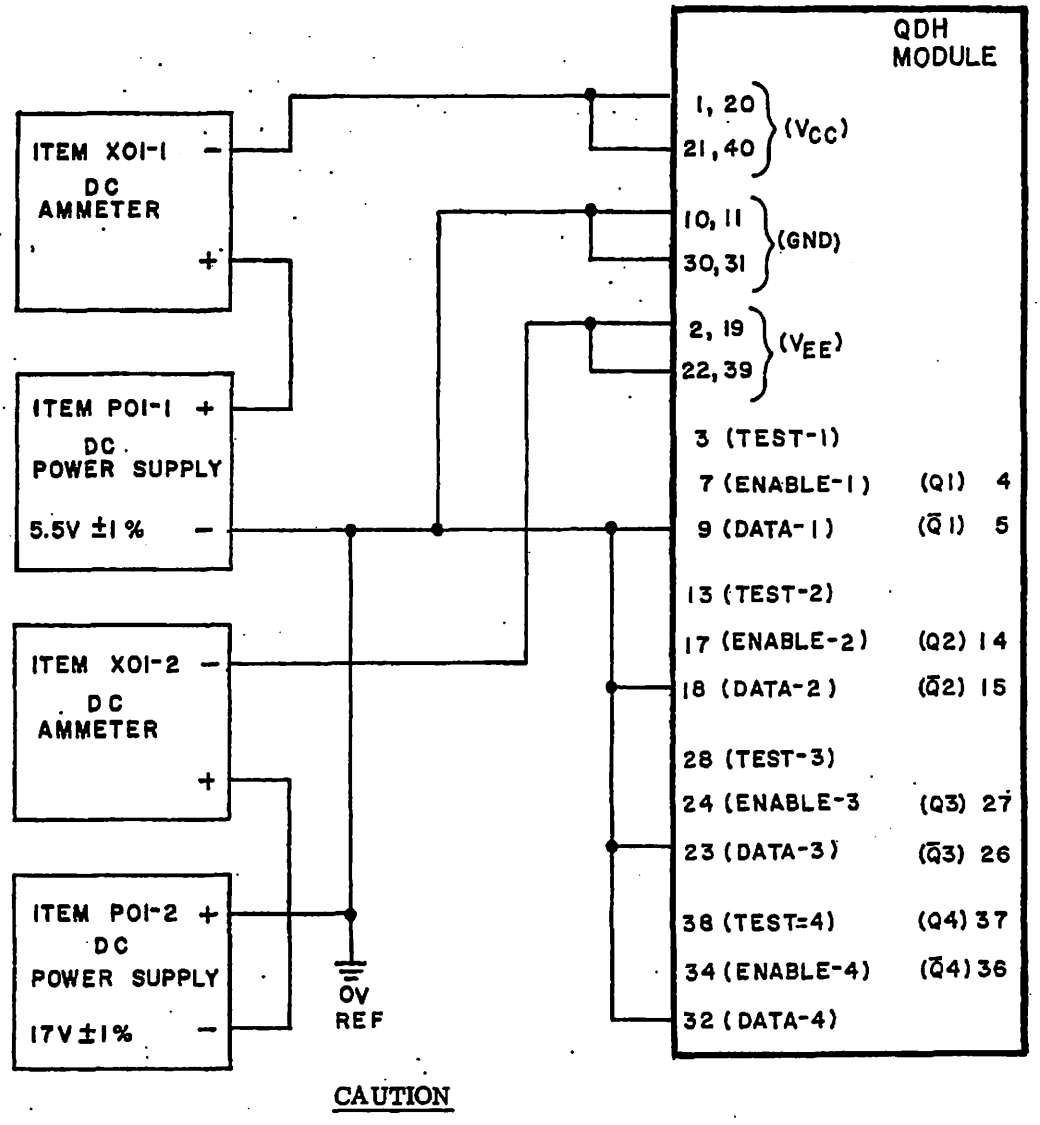
FIGURE 11. QDH logic load circuit.

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*TABLE IX. Power supply current (5 V and -15 V) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
K	Power supply current (5 V)	30	70	mA	12
L	Power supply current (-15 V)	-5	-20	mA	12

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

* NOTE; Since items XOI-1 and XOI-2 are not used simultaneously, only one item XOI is required.

FIGURE 12. Power supply current test circuit for table IX.

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* TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M1	Q1	Transition time (t_{TLH})	Open	7	Open	3	4	50	1,000	ns	13, 14
M2	Q2			17		13	14				
M3	Q3			24		28	27				
M4	Q4			34		38	37				
M5	$\overline{Q}1$			7		3	5				
M6	$\overline{Q}2$			17		13	15				
M7	$\overline{Q}3$			24		28	26				
M8	$\overline{Q}4$			34		38	36				
N1	Q1	Transition time (t_{THL})		7		3	4	5	25		
N2	Q2			17		13	14				
N3	Q3			24		28	27				
N4	Q4			34		38	37				
N5	$\overline{Q}1$			7		3	5				
N6	$\overline{Q}2$			17		13	15				
N7	$\overline{Q}3$			24		28	26				
N8	$\overline{Q}4$			34		38	36				

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* TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

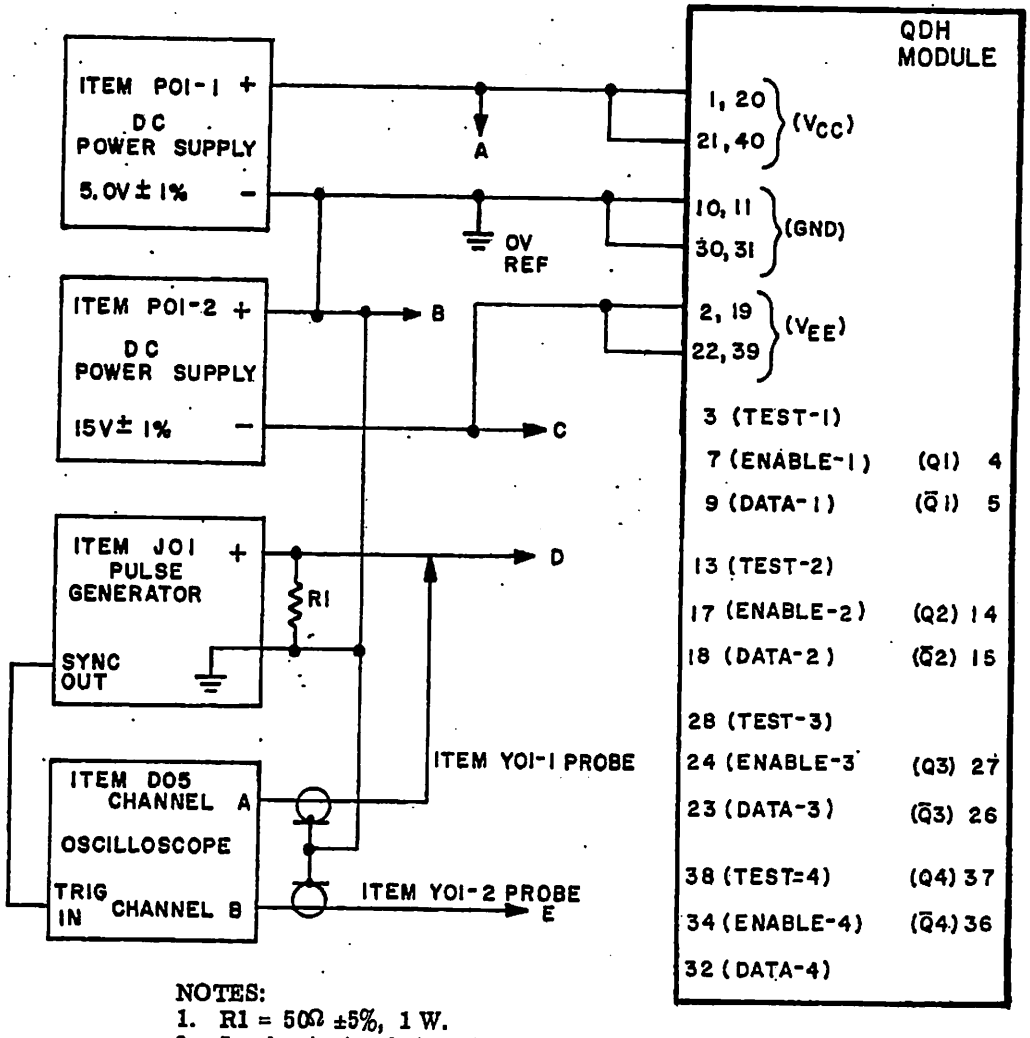
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	TEST-1 to Q1 Propagation delay time (t_{PHL})	Open	7	Open	3	4	6	40	ns	13,14
P2	TEST-2 to Q2		17		13	14				
P3	TEST-3 to Q3		24		28	27				
P4	TEST-4 to Q4		34		38	37				
P5	TEST-1 to $\overline{Q}$1		7		3	5	10	110		
P6	TEST-2 to $\overline{Q}$2		17		13	15				
P7	TEST-3 to $\overline{Q}$3		24		28	26				
P8	TEST-4 to $\overline{Q}$4		34		38	36				
P9	ENABLE-1 to Q1		3	9	7	4	6	40		
P10	ENABLE-2 to Q2		13	18	17	14				
P11	ENABLE-3 to Q3		28	23	24	27				
P12	ENABLE-4 to Q4		38	32	34	37				
P13	ENABLE-1 to $\overline{Q}$1		3	9	7	5	10	110		
P14	ENABLE-2 to $\overline{Q}$2		13	18	17	15				
P15	ENABLE-3 to $\overline{Q}$3		28	23	24	26				
P16	ENABLE-4 to $\overline{Q}$4		38	32	34	36				

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*TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	TEST-1 to Q1 Propagation delay time (t_{PLH})	Open	7	Open	3	4	10	80	ns	13,14
Q2	TEST-2 to Q2		17		13	14				
Q3	TEST-3 to Q3		24		28	27				
Q4	TEST-4 to Q4		34		38	37				
Q5	TEST-1 to $\overline{Q1}$		7		3	5		110		
Q6	TEST-2 to $\overline{Q2}$		17		13	15				
Q7	TEST-3 to $\overline{Q3}$		24		28	26				
Q8	TEST-4 to $\overline{Q4}$		34		38	36				
Q9	ENABLE-1 to Q1		3	9	7	4		80		
Q10	ENABLE-2 to Q2		13	18	17	14				
Q11	ENABLE-3 to Q3		28	23	24	27				
Q12	ENABLE-4 to Q4		38	32	34	37				
Q13	ENABLE-1 to $\overline{Q1}$		3	9	7	5		110		
Q14	ENABLE-2 to $\overline{Q2}$		13	18	17	15				
Q15	ENABLE-3 to $\overline{Q3}$		28	23	24	26				
Q16	ENABLE-4 to $\overline{Q4}$		38	32	34	36				

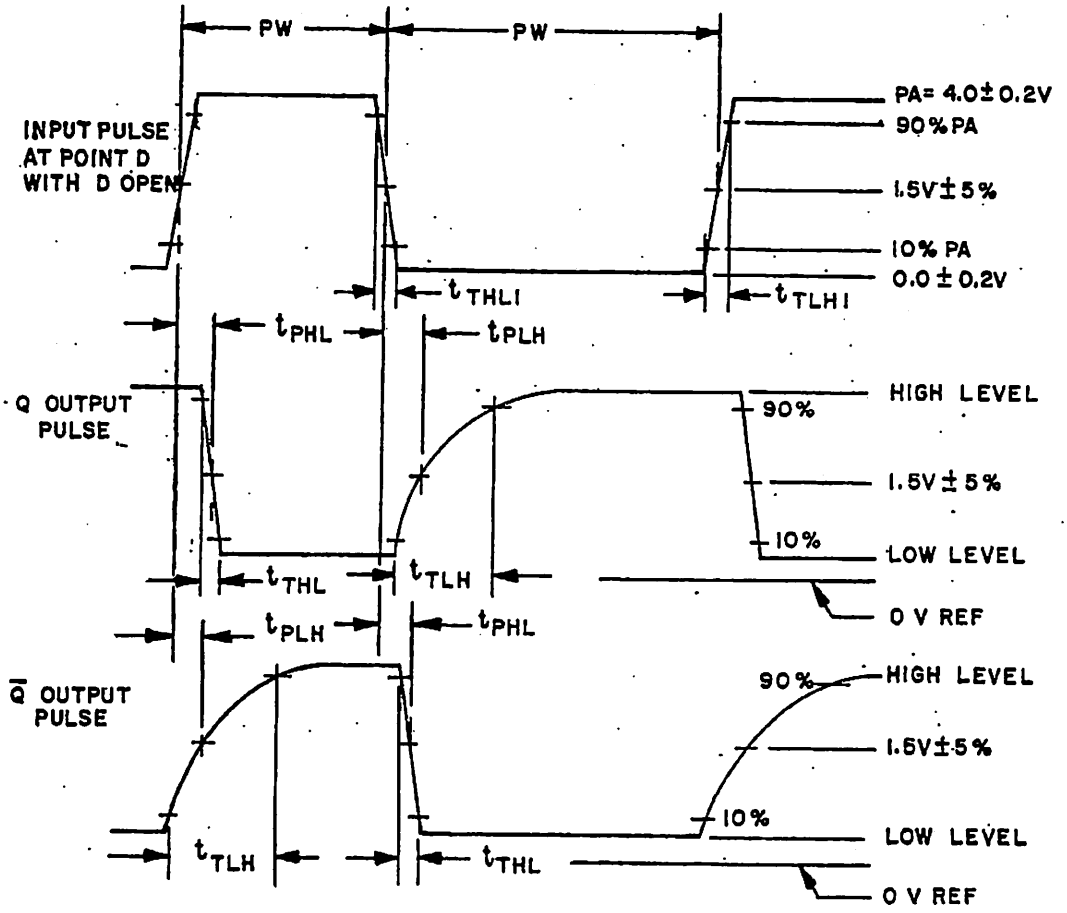
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load outputs of circuit under test with loads as specified on figure 11.

* FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for table X.

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NOTES:
1. Input pulse parameters:
   PW = 6.5 ±0.5 µs.
   $t_{TLH1} = t_{THL1}$ = 20 ±2 ns.
2. Before making measurements, verify that the output high level is 3.8 V minimum and the output low level is 0.4 V maximum.
3. All voltages are referenced to ground.

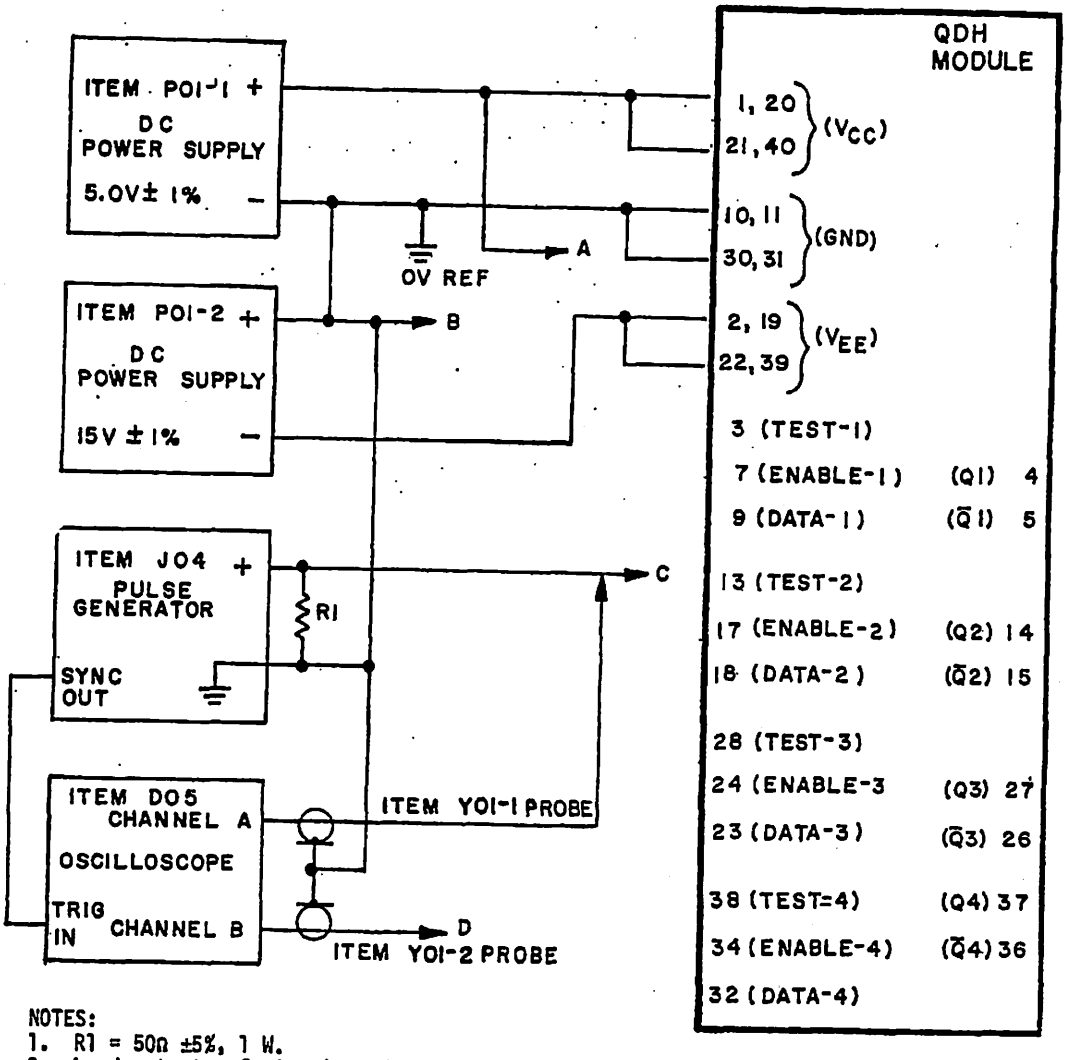
* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table X.

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*TABLE XI. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	DATA-1 to Q1 Propagation delay time (t_{PHL})	7	3	9	4	1.0	3.0	µs	15,16
R2	DATA-2 to Q2	17	13	18	14				
R3	DATA-3 to Q3	24	28	23	27				
R4	DATA-4 to Q4	34	38	32	37				
R5	DATA-1 to $\overline{Q1}$	7	3	9	5				
R6	DATA-2 to $\overline{Q2}$	17	13	18	15				
R7	DATA-3 to $\overline{Q3}$	24	28	23	26				
R8	DATA-4 to $\overline{Q4}$	34	38	32	36				
S1	DATA-1 to Q1 Propagation delay time (t_{PLH})	7	3	9	4				
S2	DATA-2 to Q2	17	13	18	14				
S3	DATA-3 to Q3	24	28	23	27				
S4	DATA-4 to Q4	34	38	32	37				
S5	DATA-1 to $\overline{Q1}$	7	3	9	5				
S6	DATA-2 to $\overline{Q2}$	17	13	18	15				
S7	DATA-3 to $\overline{Q3}$	24	28	23	26				
S8	DATA-4 to $\overline{Q4}$	34	38	32	36				

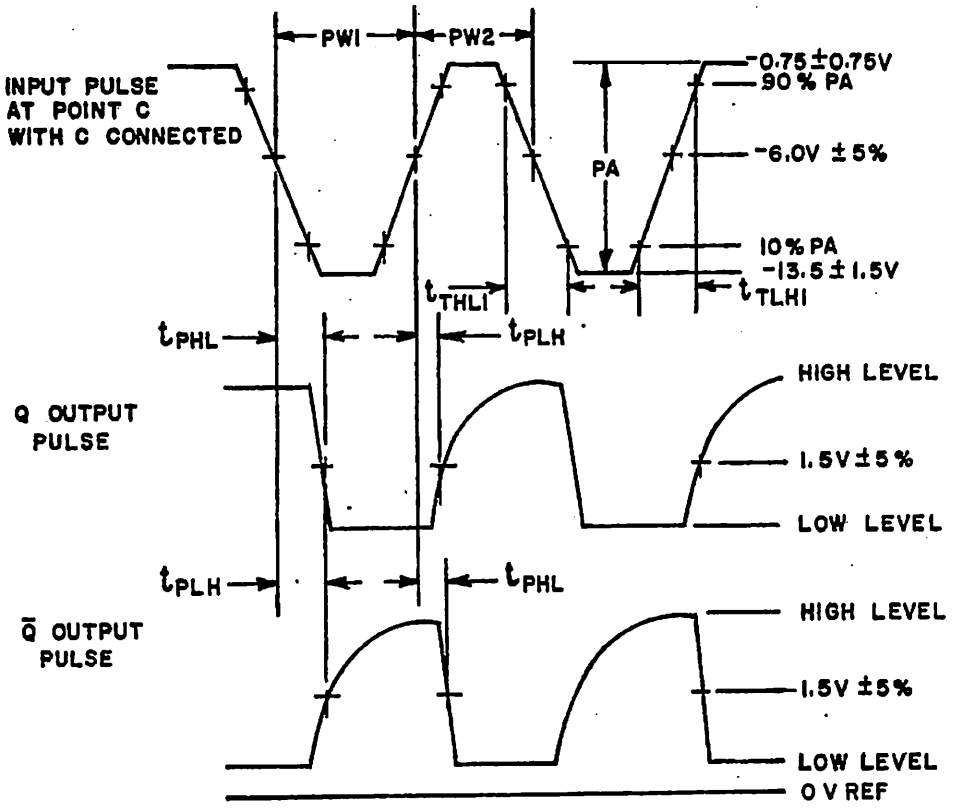
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load outputs of circuit under test with loads as specified on figure 11.

* FIGURE 15. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 3 \pm 0.5$ µs.
   PW1 = PW2 = 6.5 ±0.5 µs.
2. Before making measurement, verify that the output high level is 3.8 V minimum and the output low level is 0.4 V maximum.
3. All voltage levels are referenced to ground.

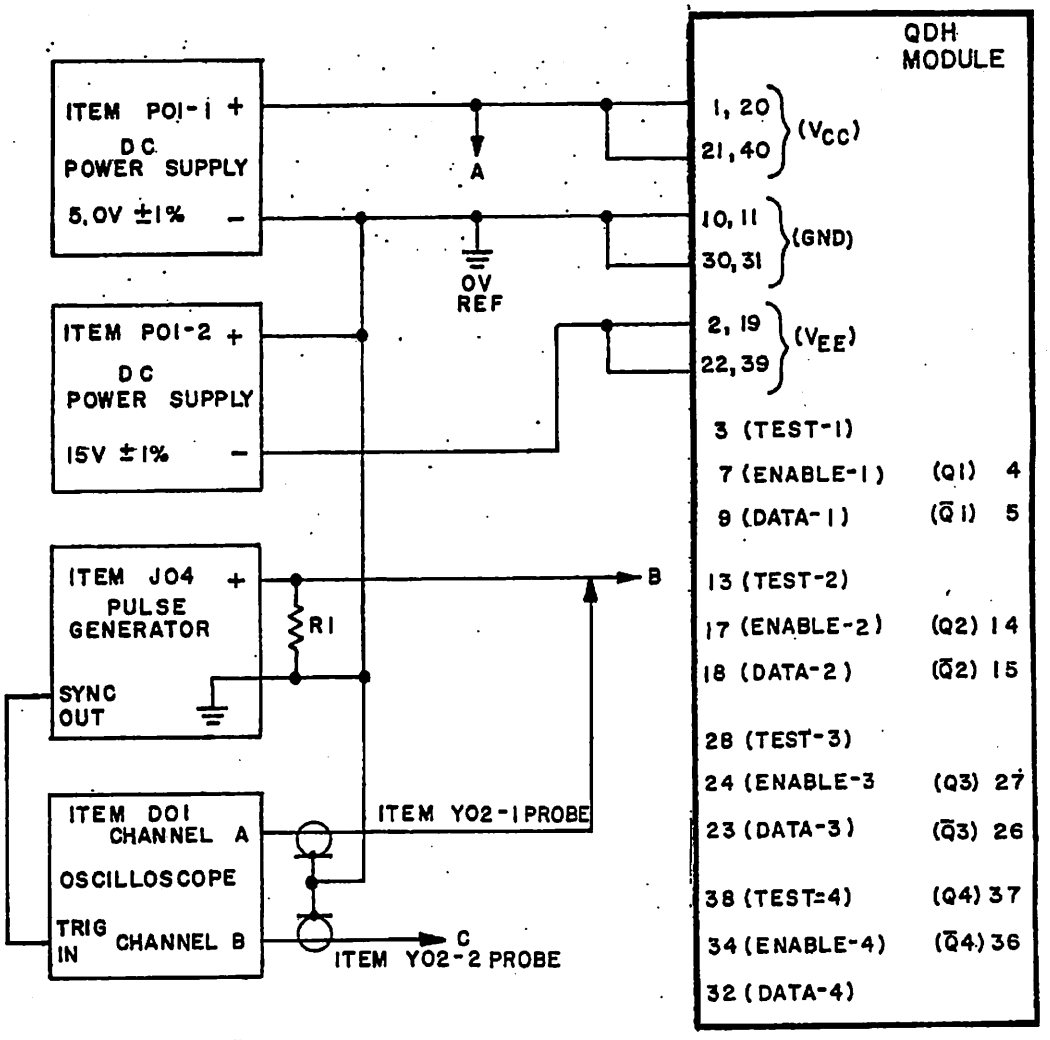
★ FIGURE 16. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XI.

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* TABLE XII. Maximum frequency check (measuring voltage level) and noise rejection (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	Maximum frequency check (measuring voltage level)	7	9	4	3.4	5.0	V	17,18
T2	↓	17	18	14	↓	↓	↓	↓
T3		24	23	27				
T4		34	32	37				
U1	Noise rejection, low level	7	9	4	0.0	0.4		17,19
U2	↓	17	18	14	↓	↓		↓
U3		24	23	27				
U4		34	32	37				
V1	Noise rejection, high level	7	9	4		0.8		17,20
V2	↓	17	18	14	↓	↓	↓	↓
V3		24	23	27				
V4		34	32	37				

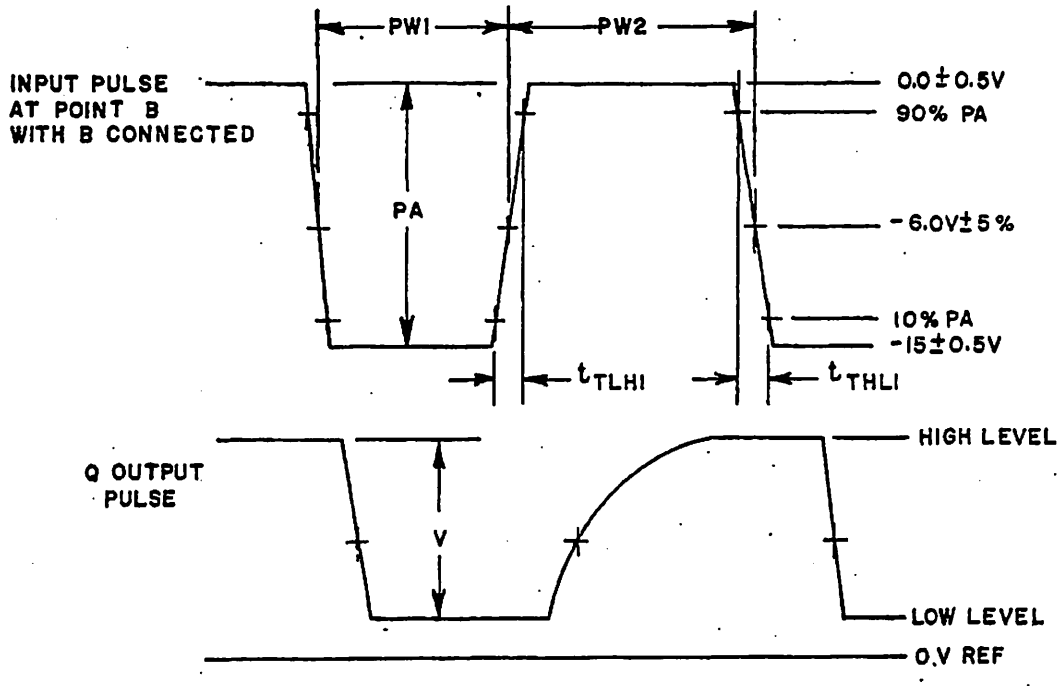
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load outputs of circuit under test with loads as specified on figure 11.

* FIGURE 17. Maximum frequency check (measuring voltage level) and noise rejection (low-level and high-level) test circuit for table XII.

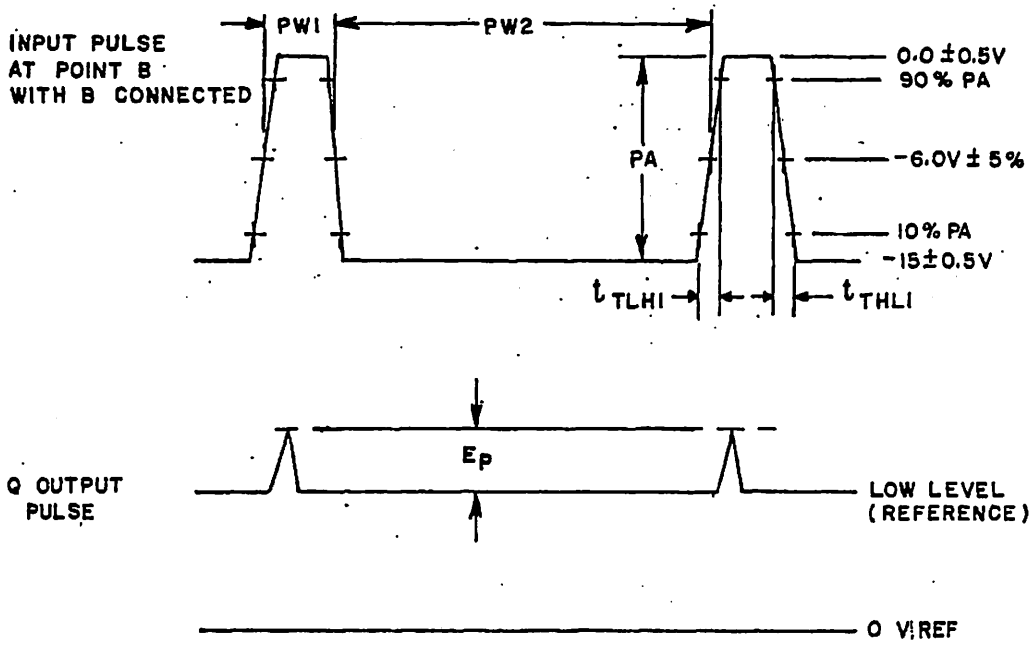
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 20 \pm 2$ ns.
   PW1 = PW2 = 2.0 ±0.1 µs.
2. All voltage levels are referenced to ground.
3. The value V is measured from the output low level to the output high level.

* FIGURE 18. Maximum frequency check (measuring voltage level) test circuit waveforms for table XII.

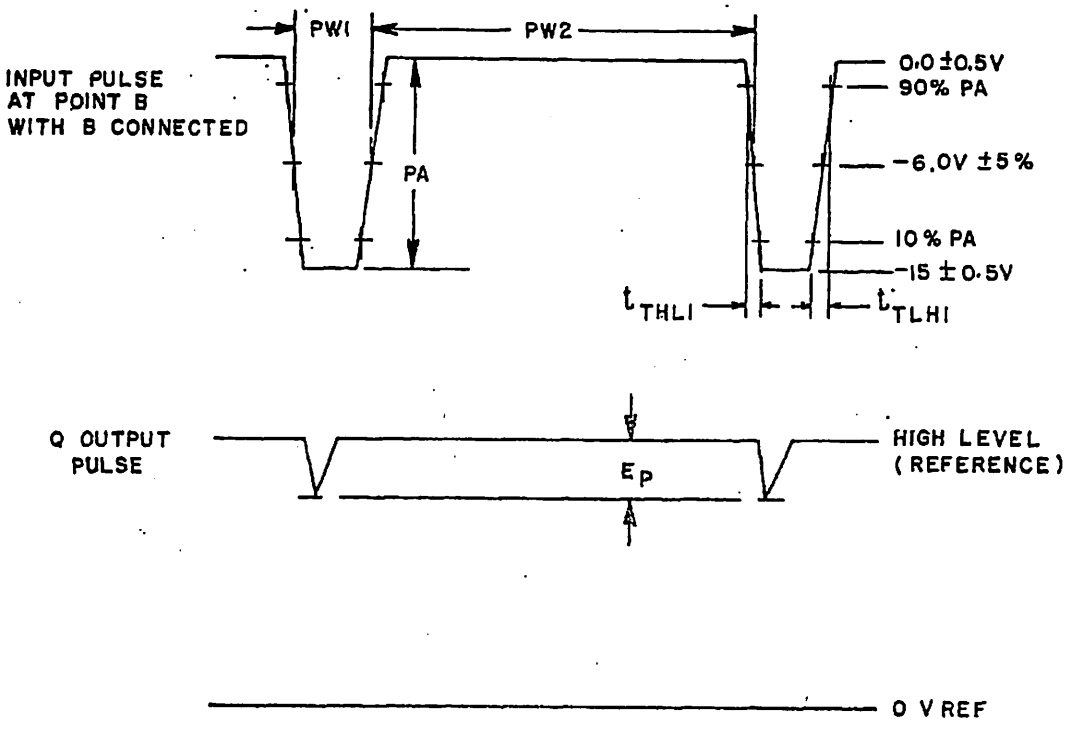
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Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 20 \pm 2$ ns.
   PW1 = 1.0 +0.1, -0.0 µs.
   PW2 = 16.1 ±1.0 µs.
2. All voltage levels are referenced to ground.
3. The value E_p is measured from the low level reference to the most positive peak of the low level output.

* FIGURE 19. Noise rejection (low-level) test circuit waveforms for table XII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 20 \pm 2$ ns.
   PW1 = $1.0 \, ^{+0.1}_{-0.0}$ µs.
   PW2 = 16.1 ±1.0 µs.
2. All voltages are referenced to ground.
3. The value E_p is measured from the high level reference to the most negative peak of the high level output.

* FIGURE 20. Noise rejection (high-level) test circuit waveforms for table XII.

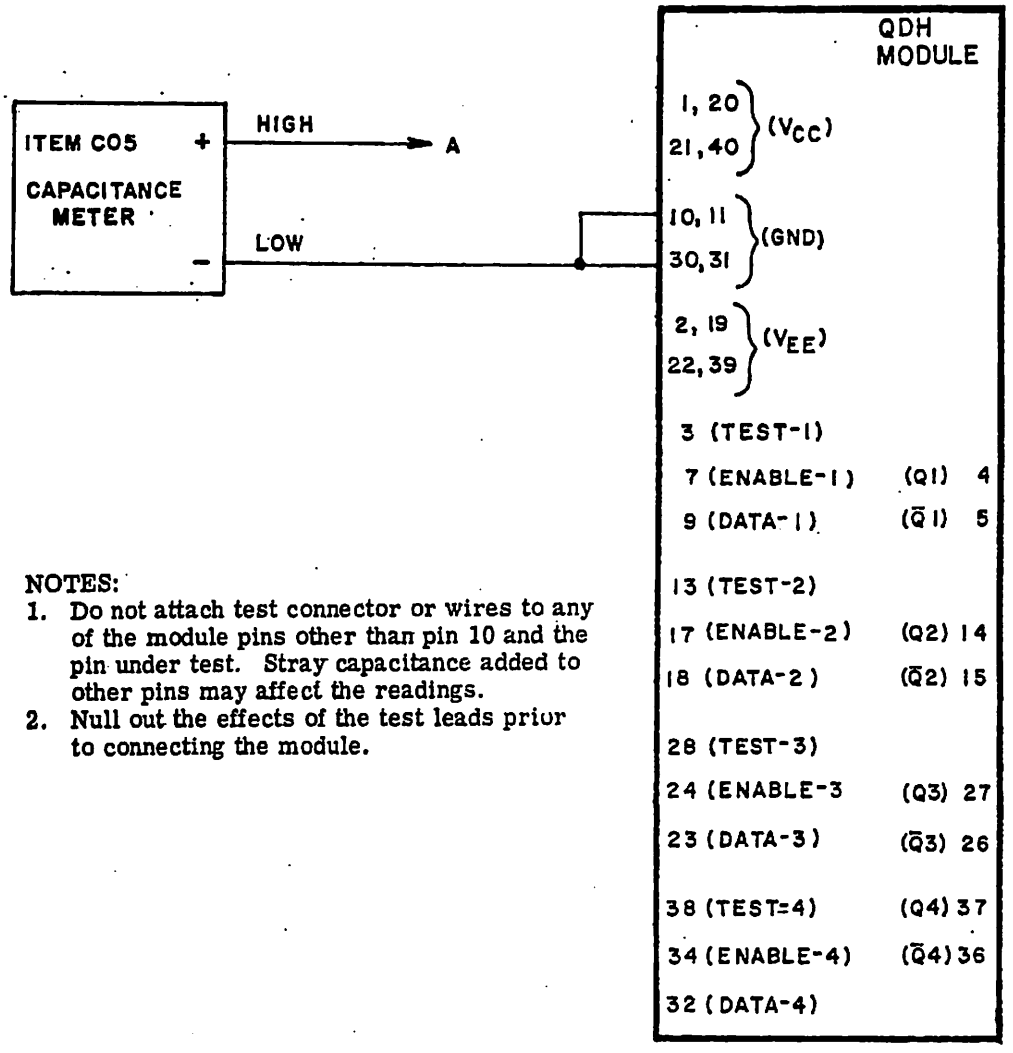
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TABLE XIII. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W1	TEST-1	3	2	15	pF	21
W2	TEST-2	13				
W3	TEST-3	28				
W4	TEST-4	38				
W5	ENABLE-1	7	6	30		
W6	ENABLE-2	17				
W7	ENABLE-3	24				
W8	ENABLE-4	34				

Input capacitance

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 21. Input capacitance test circuit for table XIII.

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